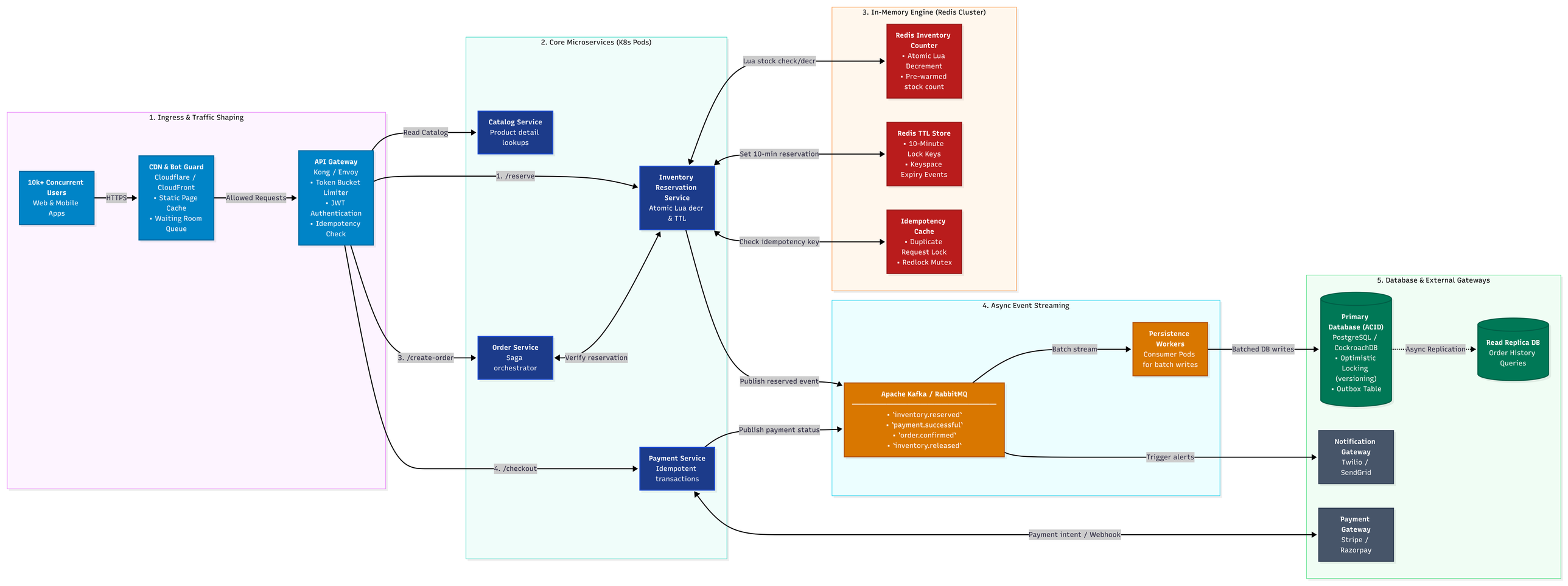

1. Ingress & Traffic Shaping
10k+ Concurrent Users
Web & Mobile Apps
HTTPS
CDN & Bot Guard
Cloudflare / CloudFront
• Static Page Cache
• Waiting Room Queue
Allowed Requests
API Gateway
Kong / Envoy
• Token Bucket Limiter
• JWT Authentication
• Idempotency Check
2. Core Microservices (K8s Pods)
Read Catalog
Catalog Service
Product detail lookups
1. /reserve
Inventory Reservation Service
Atomic Lua decr & TTL
3. /create-order
Order Service
Saga orchestrator
Verify reservation
4. /checkout
Payment Service
Idempotent transactions
3. In-Memory Engine (Redis Cluster)
Lua stock check/decr
Redis Inventory Counter
• Atomic Lua Decrement
• Pre-warmed stock count
Set 10-min reservation
Redis TTL Store
• 10-Minute Lock Keys
• Keyspace Expiry Events
Check idempotency key
Idempotency Cache
• Duplicate Request Lock
• Redlock Mutex
4. Async Event Streaming
Publish reserved event
Publish payment status
Apache Kafka / RabbitMQ
• `inventory.reserved`
• `payment.successful`
• `order.confirmed`
• `inventory.released`
Batch stream
Persistence Workers
Consumer Pods for batch writes
Trigger alerts
5. Database & External Gateways
Batched DB writes
Primary Database (ACID)
PostgreSQL / CockroachDB
• Optimistic Locking (versioning)
• Outbox Table
Async Replication
Read Replica DB
Order History Queries
Notification Gateway
Twilio / SendGrid
Payment Gateway
Stripe / Razorpay
Payment intent / Webhook